FREVER
MWALE

University of Malawi
CHANCELLOR COLLEGE
Mathematical Sciences Department

STA202 Examinations – Foundations of Probability & Statistics

Date: 2nd February 2009 **Time allowed:** Three hours

Instructions:

(1) This is a closed book exam where you are expected to do each question alone without reference to some other notes or communication.

(2) Green booklets containing statistical tables are provided.

(3) Non-programmable calculators are allowed. However, mobile phones are not allowed.

(4) Start every new question on a new page. The order that you do the questions does not matter.

(5) Show all working or reasoning as most marks are allocated to the method. Marks for each part appear in square brackets. Each question has 40 marks.

(6) Attempt **ANY three** questions of your choice.

Question 1 [40 marks]

(a) An Excel command "=ROUND(1000*RAND(),0)" generates an integer. Find, **without using a calculator,** the probability that the generated integer is

(i) 123; **[3 marks]**

(ii) 1007; and **[1 mark]**

(iii) not divisible by 3. **[5 marks]**

(b) Given that the probability it will rain today is 0.75; that it will be windy today is 0.55 and that it will either be rainy or windy today is 0.99; calculate the probability that it will both be rainy and windy today. **[6 marks]**

(c) If the probability that I will get this question right having read my class notes well is 0.97 and that I have read my class notes well and will get this question right is 0.54; calculate the probability that I have read my class notes well. **[5 marks]**

(d) Let E and F be two independent events. Prove, **from first principles,** that complementary events E^C and F^C are also independent. **[8 marks]**

(e) Suppose epidemiologically, city records show that the likelihood that a resident dies due to some illness is 98%; that a resident dies due to other sources other than illness is 5% and that the chances that one has some illness is 25%. Calculate the probability that a randomly chosen name from the city records will be a name of someone who already died. **[6 marks]**

(f) There are three entrances into a performance theatre. Ten per cent of the audience used entrance 1, 40% used entrance 2 and 50% used entrance 3. Given that the probabilities that one seats on the front row having used entrances 1, 2 and 3 are half; one-quarter and one-third, respectively, and that a person is randomly chosen from the front row, calculate the chances that this person used entrance 2 when coming into the theatre. **[6 marks]**

$P(x \cap y) =$

$P(x/y) = \frac{P(x \cap y)}{P(y)}$

$P(x \cup y)^c = P(x \cap y)^c$

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Question 2 [40 marks]

Identify the most appropriate type of probability function for the random variable Y and calculate the required probability in each case:

(a) $P\{Y = 4\}$ where Y is the number of tosses of a biased coin needed to get the first head, given that the probability that the coin shows head is 0.49 on each and every toss; **[4 marks]**

(b) $P\{Y > 6\}$ where Y is the number of daughters in a couple with 7 children; **[4 marks]**

(c) $P\{Y = 3\}$ where Y is the number of sons among 4 distinct randomly chosen children from a family of 8 children where 3 of them are sons; **[4 marks]**

(d) $P\{Y = 5\}$ where Y is the number of days to have the fourth day of petrol shortage at a filling station, if the probability that the station will not have petrol is 0.91 each day independent of any other day; **[4 marks]**

(e) $P\{Y = 0\}$ where Y is the number of ringing phone calls at a Porter's Lodge within 10 minutes if calls are coming randomly at a rate of 1 call per hour; **[4 marks]**

(f) $P\{Y \leq 2 \text{ or } Y \geq 7\}$ where Y can equally likely get any of the values in the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$; **[4 marks]**

(g) $P\{Y \leq 1.5\}$ where Y is the time period (in years) between two successive births of offspring from an animal that bears one offspring at a time at random times with an average rate of 2 offspring per year; **[4 marks]**

(h) $P\{\text{a passenger misses the bus}\}$ in terms of y where Y is the time a bus leaves a bus stop given that the bus is equally likely to depart at the bus stop any time between 10am and 11am and the passenger, who aims at boarding this bus, arrives at the bus stop at y minutes after 10 o'clock in the morning; $0 \leq y \leq 60$; **[4 marks]**

(i) $P\{Y \leq -1.96\}$ where $Y = (X - \mu)/\sigma$ for a normal random variable X with mean μ and variance σ^2; and **[4 marks]**

(j) $P\{Y = 1\}$ where Y is the total number of odd numbers that appear in successive tosses of a fair die until the third even number appears (taking the six faces of a die as 1, 2, 3, 4, 5, 6). **[4 marks]**